

DSA / Algorithms (Abdul Bari) - Day 24

45-day plan mapped to the Algorithms playlist

Matrix Chain Multiplication

MCM via DP - problem and program.

Watch these Abdul Bari videos

- > 4.3 Matrix Chain Multiplication - Dynamic Programming
- > 4.3.1 Matrix Chain Multiplication (Program) - Dynamic Programming

Key concepts

- Find the cheapest way to parenthesise a matrix product.
- DP over chain lengths.
- Minimise scalar multiplications.

Key idea

```
m[i][j]=min over k of m[i][k]+m[k+1][j]+cost.
```

Tasks (1.5h)

- [] Watch both videos
- [] Fill the MCM table by hand
- [] Code the DP

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0